



A Single-Dose Study to Evaluate the Relative Bioavailability, Safety, and Tolerability of Monthly Extended-Release Buprenorphine at Alternative Injection Locations in Adult Participants with Opioid Use Disorder

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Abstract

Background and Objectives Buprenorphine extended-release monthly formulation (BUP-XR, SUBLOCADE®) is approved for treatment of moderate-to-severe opioid use disorder (OUD) following subcutaneous injection in the abdomen. This open-label pharmacokinetic study assessed three alternative injection locations (upper arm, thigh, buttocks) to offer additional flexibility considering the chronic nature of the disease and patient preferences.

Methods Following stabilization on 12/3 mg/day of sublingual buprenorphine/naloxone for ≥ 7 days, participants with moderate-to-severe OUD were randomized to receive a single 300-mg BUP-XR injection in the upper arm, thigh, buttocks, or abdomen (reference). Serial blood samples were taken to measure buprenorphine plasma concentrations over 28 days and assess buprenorphine relative bioavailability. Safety evaluations included treatment-emergent adverse events and assessments of injection site pain, tenderness, erythema, induration, and swelling.

Results A total of 88 participants received a single subcutaneous injection of 300-mg BUP-XR in the upper arm ($N = 21$), thigh ($N = 23$), buttocks ($N = 22$), or abdomen ($N = 22$); 81/88 (92%) completed the study. Buprenorphine plasma exposure (area under the plasma concentration-time curve over 28 days) was comparable across injection site groups with mean buprenorphine plasma concentrations sustained at approximately 2 ng/mL (therapeutic target concentration) or above. Buprenorphine maximum plasma concentration ($C_{\max}$) was approximately 39% and 52% higher after injection in the upper arm and thigh, respectively, versus the abdomen, while comparable between buttocks and abdomen. Higher $C_{\max}$ values were not associated with an increased incidence of adverse events. Safety and injection site tolerability were comparable across injection groups.

Conclusions These pharmacokinetic and safety findings support BUP-XR injection into the upper arm, thigh, and buttocks.

Trial Registration Clinicaltrials.gov: NCT05704543

1 Introduction

Opioid use disorder (OUD) is a chronic, relapsing disease associated with significant morbidity and mortality [1]. In the USA, approximately 6.1 million people aged 12 years or older were diagnosed with OUD in 2022, and 76,508 opioid-involved overdose deaths were reported over the 12-month period ending in February 2024 [2, 3]. Medications for OUD (MOUD, e.g., buprenorphine, methadone, and naltrexone) are effective treatments and represent the standard of care when combined with psychosocial support [4]. However, various barriers prevent the broad use of these medications,

and only 15% patients in the USA initiate treatment with an MOUD [5]. Barriers to treatment include limited access to treatment centers/treatment options, the need for regular clinic visits, inadequate professional education and training, fear of diversion, and the stigma associated with treatment/OUD [6–8].

Buprenorphine is a partial agonist at the mu-opioid receptor (μ -OR) and as such presents a reduced risk of respiratory depression, which is a potential advantage over full agonists such as methadone [4, 9]. While buprenorphine is most commonly dispensed to patients as a take-home transmucosal (buccal, sublingual) medication for daily administration [4, 10], it has been approved in the form of weekly or monthly injectable formulations [11] that may offer more convenience and address some of the

Key Points

Pharmacokinetic and safety results support the use of alternative injection sites (upper arm, thigh, and buttocks) for monthly subcutaneous administration of extended-release buprenorphine (BUP-XR) in participants with moderate-to-severe opioid use disorder.

BUP-XR achieved and sustained therapeutic buprenorphine plasma concentrations over the monthly dosing interval for all injection sites, with comparable buprenorphine plasma exposure and no clinically meaningful differences in safety or tolerability across injection site groups.

The higher peak buprenorphine plasma concentrations measured after subcutaneous injection of BUP-XR in the upper arm and thigh did not translate into an increased incidence of adverse events.

existing barriers to MOUD treatment. Extended-release buprenorphine (BUP-XR, SUBLOCADE®) was the first monthly subcutaneous buprenorphine injection approved in the USA [12] for the treatment of moderate-to-severe OUD and is also approved in Canada, Australia, and several countries in Europe.

BUP-XR was designed to provide sustained exposure of buprenorphine over the entire monthly dosing interval and to deliver target therapeutic plasma concentrations of ≥ 2 ng/mL (corresponding to $\geq 70\%$ μ -OR occupancy), which are necessary to block the subjective effects of exogenous opioids (i.e., opioid blockade) [13–15]. The recommended BUP-XR dosing regimen starts with two monthly injections of BUP-XR 300 mg to rapidly achieve and sustain therapeutic target levels of ≥ 2 ng/mL [12, 13, 16]. To maintain those therapeutic concentrations, the recommended maintenance dose of BUP-XR is 100 mg monthly. This maintenance dose may be increased to 300 mg monthly for patients who tolerate the 100 mg dose but do not demonstrate a satisfactory clinical response and may benefit from higher buprenorphine plasma concentrations in the range of 5–6 ng/mL [12, 13, 16, 17].

The clinical efficacy of BUP-XR was demonstrated in a double-blind, placebo-controlled study conducted in adults with moderate or severe OUD. In that study, treatment with BUP-XR for up to 6 months significantly increased abstinence from illicit opioids, improved treatment retention, and reduced opioid craving and withdrawal symptoms [14]. BUP-XR treatment was also shown to significantly improve health-related quality-of-life measures in

comparison with placebo [18]. Additional studies, including long-term safety studies over 18 months and real-world analyses, confirmed BUP-XR safety and tolerability and its sustained efficacy [19–22]. Notably, a randomized, controlled, open-label, 6-month trial at multiple community-based treatment centers in England and Scotland demonstrated the superiority of BUP-XR over standard-of-care treatment (i.e., transmucosal buprenorphine or oral methadone), with a greater abstinence from opioids, higher likelihood of early OUD remission, longer retention in treatment, and reduced or lack of opioid craving [22]. In all these studies, BUP-XR was administered by subcutaneous injection in the abdomen.

Because BUP-XR has been approved for subcutaneous injection in the abdominal region only [12], the present study investigated three additional injection locations (thigh, upper arm, buttocks) to account for patients' preferences and facilitate the chronic use of this long-acting injectable product. The primary objective of this study was to assess the relative bioavailability of buprenorphine following a single subcutaneous BUP-XR injection into those alternative sites compared with the abdomen in participants with OUD. A secondary objective was to evaluate the safety and tolerability of BUP-XR injection at those alternative sites.

2 Methods

2.1 Study Design

This multicenter, randomized, open-label, single-dose, parallel-group study (NCT05704543) was conducted in participants with moderate-to-severe OUD at six centers in the USA from March to November 2023. It was conducted in accordance with the principles and requirements of the Declaration of Helsinki and the Council for International Organizations Medical Sciences International Ethical Guidelines. An independent Institutional Review Board reviewed and approved the protocol in accordance with the International Council on Harmonisation/Good Clinical Practice guidelines.

All participants provided written informed consent before any protocol-related procedures commenced. Eligible participants were aged 18–65 years with a body mass index (BMI) ≥ 18.0 to ≤ 33.0 kg/m². They met the Diagnostic and Statistical Manual of Mental Disorders, fifth edition (DSM-5) criteria for moderate or severe OUD [23] or had documented history of moderate to severe OUD. Participants were either on MOUD or were new to treatment and seeking MOUD (details in Supplementary Materials). Key exclusion criteria were a current diagnosis, other than OUD, requiring chronic opioid treatment;

concurrent primary substance use disorder, as defined by DSM-5 criteria [23], other than opioid, tobacco, cannabis, or mild-to-moderate alcohol use disorders; target areas unsuitable for subcutaneous (SC) injections or evaluation of injection sites; and concurrent or prior treatment with any long-acting injectable buprenorphine product in the past 3 years prior to consent, or treatment with depot naltrexone within the 3 months prior to consent.

Following a screening period of up to 30 days, all eligible participants entered a run-in period during which they were stabilized on 12/3 mg of buprenorphine/naloxone SL film (hereafter referenced as SL BUP) once daily for a minimum of 7 days and a maximum of 14 days before BUP-XR injection (Day -14 to Day -1, Supplementary Fig. 1). Both SL BUP (SUBOXONE®) [10] and BUP-XR (SUBLOCADE®) [12] were provided by Indivior Inc., North Chesterfield, VA. Participants completed SL BUP stabilization in an inpatient or outpatient setting and received at least the final 3 days of SL BUP at specified times in the residential facility. On Day 1, eligible participants were randomized (via an Interactive Web Response System using a central, blocked, stratified randomization schedule) to receive a single 300-mg subcutaneous BUP-XR injection in their abdomen (reference injection site) or in one of the following alternative test locations: back of upper arm, buttocks, or thigh. On Day 3, participants were discharged from the clinic and returned to the clinic for outpatient follow-up visits occurring on Days 4, 6, 8, 11, 15, 18, 22, 25, and 29.

2.2 Pharmacokinetic Assessments

Blood samples of approximately 6 mL were collected for the quantification of buprenorphine and norbuprenorphine plasma concentrations at the following times relative to BUP-XR injection: on Day 1 (predose and at 1, 2, 4, 6, 8, 12, 16, and 20 h post-dose), on Day 2 (at 24, 28, 32, 36, 40, and 44 h post-dose), on Day 3 (48 h post-dose), and then on Days 4, 6, 8, 11, 15, 18, 22, 25, and 29 (as close as possible to the same time of day as BUP-XR injection on Day 1). Blood samples were also taken during the run-in period following intake of SL BUP, i.e., on Day -2 (predose) and Day -1 (predose, and 30 min and 1, 2, 4, 6, 12, and 24 h post-dose).

Blood samples were collected in a pre-labelled K₂EDTA vacutainer tube by individual venipuncture, and the actual date and time of each sample were recorded. Blood samples were then centrifuged, and the plasma transferred into 2 aliquots of approximately 1.5 mL each in pre-labelled polypropylene tubes and then stored at -20 °C until analysis. Plasma concentrations of buprenorphine and norbuprenorphine were determined using a validated method of liquid chromatography with tandem mass spectrometry [24]. Briefly, analytes and deuterated internal standards

(buprenorphine-d4 and norbuprenorphine-d3) were extracted from K₂EDTA plasma with methyl tert-butyl ether/hexane (liquid-liquid extraction). Following extraction, the organic phase was dried down under nitrogen in a water bath at 40 °C, reconstituted, and transferred to plastic injection vials. The peak area of the m/z 468.5→414.4 buprenorphine transition ion was measured against the peak area of the m/z 472.5→414.4 buprenorphine-d4 internal standard transition ion. The peak area of the m/z 414.4→340.3 norbuprenorphine transition ion was measured against the peak area of the m/z 417.4→343.2 norbuprenorphine-d3 internal standard transition ion. Quantitation was performed using separate weighted ($1/x^2$ for both analytes) linear least squares regression analyses generated from calibration standards. The assay was fully validated for linearity, selectivity, recovery, matrix effect, accuracy, precision, dilution integrity, and stability before application to the sample analysis. The calibration range was 0.020–10.0 ng/mL for both buprenorphine and norbuprenorphine. The overall accuracy and precision for quality control samples during the sample analyses were all within 7.0%. All plasma samples were analyzed within the established stability window.

The following pharmacokinetic parameters were calculated for both analytes after BUP-XR administration at each injection location: AUC_{0–28 days} (area under the plasma concentration-time curve [AUC] from time zero [Day 1] to 28 days post-dose [Day 29]), AUC_{last} (AUC until the last quantifiable plasma concentration), C_{max} (maximum observed plasma concentration), T_{max} (time to maximum observed plasma concentration), and C_{trough} (concentration on Day 29). Pharmacokinetic parameters were derived by non-compartmental analysis using PhoenixTM WinNonlin[®] software (Version 8.3.4 or higher, Certara, LP). The primary endpoints were buprenorphine AUC_{0–28 days} and C_{max}. The pharmacokinetic evaluable analysis set was defined as all randomized participants who received BUP-XR injection and provided an adequate number of post randomization pharmacokinetic samples to derive the primary endpoints. No non-compartmental analysis was conducted for drug plasma concentrations measured after SL BUP administration on Day 1; only descriptive statistics and concentration plots were generated.

2.3 Safety and Tolerability Assessments

Safety was assessed for the safety analysis set, defined as all enrolled participants who received at least one dose of SL BUP during the run-in period. Treatment-emergent adverse events (AEs) were coded to system organ class and preferred term using the Medical Dictionary for Regulatory Activities (MedDRA) version 26.1. The incidence and percentage of participants with AEs during the run-in and post injection periods were summarized by system organ class and

preferred term and separately summarized for all AEs, AEs identified as injection site reactions, serious AEs, BUP-XR-related AEs, and AEs reported as an opioid withdrawal symptom. AEs were also summarized by maximum severity and relatedness within system organ class and preferred term.

AEs identified as injection site reactions were summarized using customized MedDRA queries. Injection site pain was assessed by the participant at 1, 5, 10, 15, and 30 min post-BUP-XR injection, using a 100-mm visual analog scale (VAS) scale, where 0 represents no pain and 100 represents maximum pain. The score was measured as the length between left end of the paper to the participant mark. In addition, participants answered the following question with either a yes or no: “Are you currently experiencing any burning or stinging at the injection site?” Finally, a clinical evaluation of the injection site was performed by the study site at 10 min and 2 h after BUP-XR injection, rating pain, tenderness and erythema/redness, induration, and swelling as none (0), mild (1), moderate (2), severe (3), or potentially life-threatening (4).

Physical examinations were conducted at screening. Vital signs (including temperature, systolic and diastolic blood pressure, pulse rate, and respiratory rate after the participant was in a sitting or supine position for ≥ 3 min) were taken on Day -3 (admission to the clinic), Days 1–3 of the treatment period, and at end of treatment. Electrocardiograms (ECG) were performed at screening and on Day -3. Clinical safety laboratory values were measured at screening. Liver function tests were performed at screening, on Days -3 and 3, and at the end of treatment. Urine drug screens (UDS) were performed at screening and at the last visit/end of the study.

2.4 Statistical Analyses

No formal statistical power analysis was performed to determine the sample size of this comparative bioavailability study as no hypothesis testing was planned. On the basis of prior pharmacokinetic knowledge [16] and experience from similar clinical trials, 20 participants per group were planned to achieve at least 15 pharmacokinetic evaluable participants in each injection site group, which was deemed sufficient to meet the primary pharmacokinetic objective of the study.

Buprenorphine and norbuprenorphine pharmacokinetic parameters were summarized by injection site using descriptive statistics, including the geometric mean and coefficient of variation (CV%). $T_{\max}$ values were summarized as the median, minimum, and maximum. Other continuous variables were summarized using general descriptive statistics such as mean, standard deviations (SD) and/or CV%, median, minimum, and maximum. Categorical variables were reported as frequency counts (including number

missing) and the percentage of participants in corresponding categories. The pharmacokinetic profiles of buprenorphine and norbuprenorphine were summarized graphically by injection site, using both linear and semi-logarithmic scales.

To descriptively compare buprenorphine and norbuprenorphine pharmacokinetic parameters between injection sites, natural log-transformed $AUC_{0-28 \text{ days}}$, AUC_{last} , and $C_{\max}$ were analyzed using an analysis of variance (ANOVA) model with injection site as the independent variable. A 90% confidence interval (CI) was estimated for the ratio of population geometric means (GMR, test/reference) corresponding to exponentiated differences in least squares means. The comparisons of interest were the three alternative injection sites (test) versus abdomen (reference), namely: back of upper arm versus abdomen, buttocks versus abdomen, and thigh versus abdomen.

Treatment-emergent AEs, injection site pain VAS, and injection site grading were summarized using descriptive statistics and any differences between alternative injection locations versus the abdomen were estimated with a 95% CI. Both injection site VAS and injection site grading were tabulated according to nominal timepoints. For the analysis of AEs, missing severity and relatedness were queried until resolution. If resolution was not possible, missing severity was summarized as “severe” and missing relatedness was summarized as “related.”

3 Results

3.1 Study Participants

Of 175 participants screened, 132 were enrolled and received SL BUP film during the run-in period. Of those enrolled, 44 (33.3%) discontinued during run-in and 88 were randomized and dosed with BUP-XR (Fig. 1). The most common reason for discontinuation during run-in was withdrawal of consent (39 participants). Randomized participants were allocated to the following injection site groups: abdomen ($n = 22$), upper arm ($n = 21$), buttocks ($n = 22$), and thigh ($n = 23$). Of the 88 randomized participants who received BUP-XR injection, 81 (92%) completed the study and 7 discontinued (4 withdrew consent and 3 were lost to follow-up, Fig. 1). No one was withdrawn because of an AE during the run-in or BUP-XR treatment period. The safety analysis set and pharmacokinetic evaluable analysis set included 88 and 85 participants, respectively.

The mean (SD) age of the participant population was 41.2 (9.77) years (Table 1, range 21–64 years) and the majority of participants were white (67.0%) and male (64.8%). There were no clinically meaningful differences in the distribution of sex or race between injection site

groups. Of study participants, 73.9% reported using heroin and 53.4% reported using fentanyl (data not shown). At screening, 66 (75.0%) of the 88 participants who received BUP-XR had an opioid positive UDS.

3.2 Pharmacokinetics

After SL BUP dosing on Day – 1 (run-in), buprenorphine pharmacokinetic profiles were similar across injection site groups (Supplementary Fig. 2). Predose buprenorphine plasma concentrations measured on Day – 2, Day – 1, and Day 1 were comparable, with mean (SD) values of 1.76 (1.91), 1.57 (2.01), and 1.37 (1.39) ng/mL, respectively, indicating that steady-state had been achieved. Buprenorphine pharmacokinetic profiles following BUP-XR injection are shown in Fig. 2 per injection site group. Mean buprenorphine plasma concentrations increased rapidly after each BUP-XR injection and tended to be higher following injection in the thigh and upper arm compared with injection in the buttocks and abdomen over the first 6 h post-injection. After that time, buprenorphine plasma concentrations were comparable across injection site groups, decreasing gradually to a plateau and maintained at target therapeutic levels of approximately 2 ng/mL or higher. Norbuprenorphine pharmacokinetic profiles were also comparable across injection site groups, as shown in Supplementary Fig. 3.

Table 2 presents geometric mean ratios (GMRs) with their 90% CIs for comparison of buprenorphine and norbuprenorphine pharmacokinetic parameters following injection

in the upper arm, buttock or thigh, compared with the reference abdominal location. Buprenorphine plasma exposure was comparable across the four injection site groups, with GMRs ranging from 95.89 to 114.11% for $AUC_{0-28 \text{ days}}$ and from 96.17 to 115.47% for AUC_{last} (Table 2). Buprenorphine peak exposure (C_{max}) was similar following injection in the buttocks versus the abdomen (GMR 107.47%). However, buprenorphine C_{max} was higher following injection in the upper arm and thigh compared with the abdomen (GMR 138.71% and 152.27%, respectively). These higher buprenorphine C_{max} were associated with an earlier median T_{max} of 4.0 h and 7.0 h for the thigh and upper arm respectively, compared with 10 h and 16 h for the buttocks and abdomen (Supplementary Table 1). It should be noted though that individual pharmacokinetic parameter values were largely overlapping across injection site groups, with only a few individuals having much higher C_{max} and $AUC_{0-28 \text{ days}}$ following BUP-XR injection in the thigh and upper arm (see Supplementary Fig. 4). For norbuprenorphine, Table 2 presents similar plasma exposures ($AUC_{0-28 \text{ days}}$, AUC_{last}) and peak exposures (C_{max}) across injection locations, with GMRs ranging from 89.63 to 98.67% for AUCs and from 97.73 and 99.80% for C_{max} .

3.3 Safety and tolerability

During the run-in period, of 132 participants who received SL BUP, 26 (19.7%) had at least 1 AE; 10 (7.6%) had AEs of opioid withdrawal symptoms; and 10 (7.6%) had SL

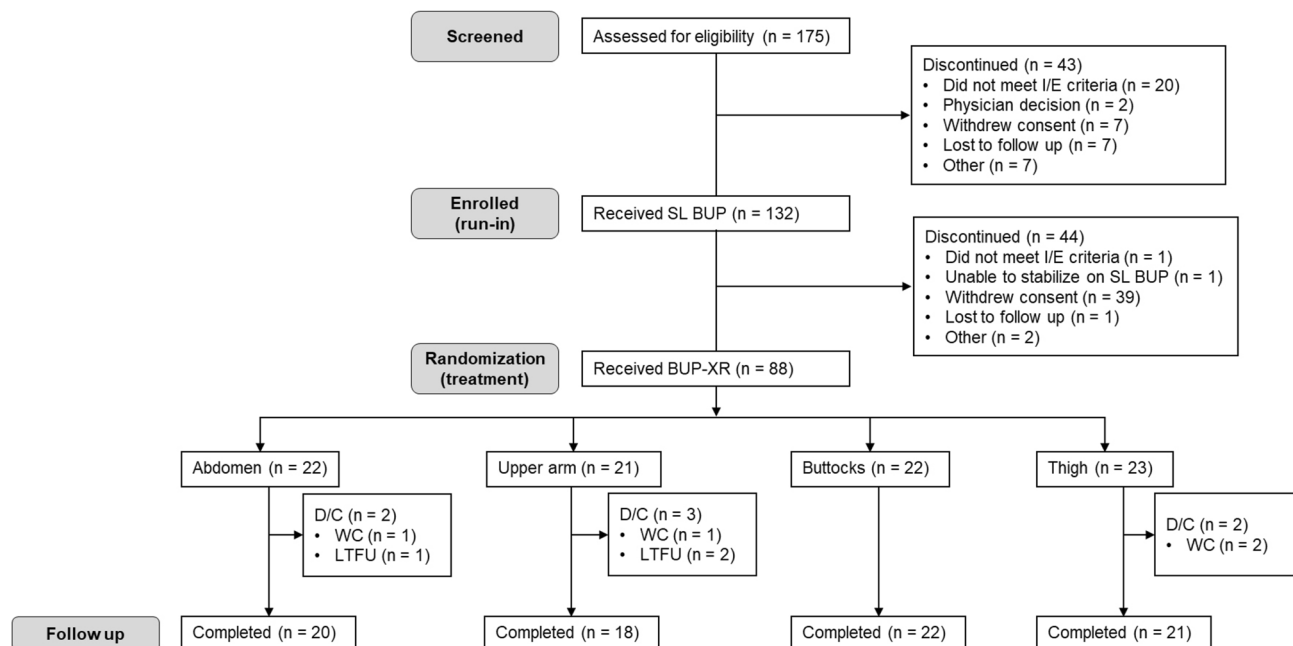


Fig. 1 Disposition of participants. BUP-XR, extended-release buprenorphine; D/C, discontinued; I/E, inclusion/exclusion criteria; LTFU, lost to follow-up; SL BUP, buprenorphine/naloxone sublingual film; WC, withdrew consent

Table 1 Demographics and baseline characteristics

Parameter	Injection location				Total <i>N</i> = 88
	Abdomen <i>N</i> = 22	Upper arm <i>N</i> = 21	Buttocks <i>N</i> = 22	Thigh <i>N</i> = 23	
<i>Age, years</i>					
Mean ± SD	39.0 ± 10.35	43.9 ± 9.61	45.6 ± 8.94	36.7 ± 7.89	41.2 ± 9.77
<i>Sex, n (%)</i>					
Male	15 (68.2)	15 (71.4)	15 (68.2)	12 (52.2)	57 (64.8)
Female	7 (31.8)	6 (28.6)	7 (31.8)	11 (47.8)	31 (35.2)
<i>Race, n (%)</i>					
Black or African American	7 (31.8)	6 (28.6)	8 (36.4)	4 (17.4)	25 (28.4)
White	14 (63.6)	14 (66.7)	13 (59.1)	18 (78.3)	59 (67.0)
American Indian, Alaska Native	0 (0.0)	1 (4.8)	0 (0.0)	0 (0.0)	1 (1.1)
Other	1 (4.5)	0 (0.0)	1 (4.5)	1 (4.3)	3 (3.4)
<i>Baseline BMI, kg/m²</i>					
Mean ± SD	26.87 ± 4.698	25.95 ± 5.011	26.06 ± 4.259	26.30 ± 4.500	26.30 ± 4.552
Median	26.70	26.50	25.10	25.10	25.80
<i>Lifetime opioid use, years^a</i>					
Mean ± SD	20.8 ± 13.55	21.0 ± 9.94	18.9 ± 9.65	18.7 ± 10.34	19.8 ± 10.85
Median	20.0	21.0	18.0	19.0	20.0

BMI, body mass index; SD, standard deviation

^aLifetime use was calculated as: the latest stop date year – the earliest start date year + 1. If the use was ongoing, then the year of the screening visit was used as the stop date year. Shown for the safety analysis set

BUP-related AEs. None had serious AEs, severe AEs, or AEs leading to SL BUP discontinuation or death. During the BUP-XR treatment period, the safety profile at alternative injection locations was comparable to that of the abdomen (Table 3). All AEs were mild or moderate in severity and no severe or serious AEs were reported. There were no AEs of opioid withdrawal symptoms reported after BUP-XR injection in any group.

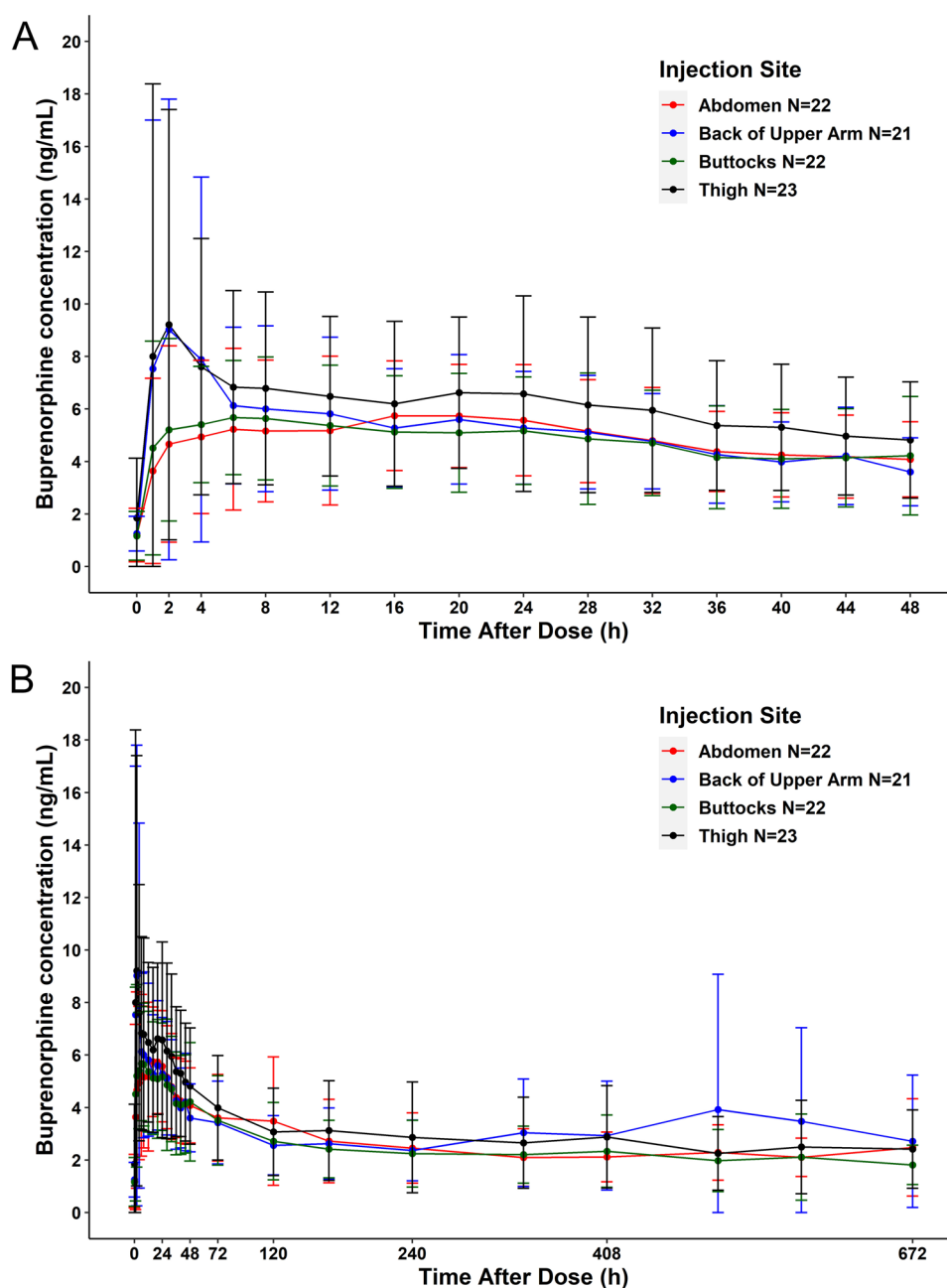
Most AEs reported for > 1 participant were related to the injection (injection site erythema, induration, pain, and swelling, Supplementary Table 2). However, there were no significant differences in the incidence of injection site reaction AEs between the test injection sites and the reference abdominal site (Table 3). In addition, there were no statistically significant differences in injection site gradings assessed at 10 min and 2 h post-injection between test injection sites and the abdomen (Supplementary Tables 3 and 4), with the exception of pain at 2 h post-injection, which was slightly higher in the upper arm versus abdomen (difference 0.19; 95% CI 0.02, 0.36, Supplementary Table 4); however, corresponding grades were all mild or none. Across injection site groups, most participants had grades of 0 (none) or 1 (mild) on all items (pain, tenderness, erythema, induration, and swelling, Supplementary Tables 3 and 4). Grades of 2 (moderate) were only reported at 10 min after BUP-XR injection for pain (2–3 participants in each injection group) and tenderness (1 and 2 participants in the thigh and

abdomen injection groups, respectively). In the majority of the participants, injection site pain/burning or stinging decreased over time (1–30 min post-dose), as assessed in a separate yes/no question (data not shown).

Participant-rated injection site pain VAS scores were not statistically different between test injection sites and the abdomen. The highest mean injection site pain VAS scores, recorded 1 min after the BUP-XR injection, were not statistically different across injection site groups, ranging from 27.4 (buttocks) to 37.5 (abdomen, Fig. 3). By 15 min post-injection, mean pain scores decreased to < 10 in all injection site groups.

Overall, the BUP-XR injection was well tolerated across all injection sites. There were no new safety signals detected relative to the known safety profile of BUP-XR with the abdominal injection. Vital signs remained within the normal reference range during the study with a few minor exceptions that were not of clinical importance. There were no clinically meaningful changes in ECGs. In addition, no new safety signals relative to the known safety profile of BUP-XR were detected on the basis of evaluation of hematology, clinical chemistry, urinalysis, urine drug screen, coagulation, and serology laboratory results. There were no deaths or serious AEs reported during this study. No participant had a serious adverse event potentially pertaining to liver dysfunction and no cases of Hy's Law were reported.

Fig. 2 Buprenorphine plasma concentration-time profiles (mean $\pm$ SD) after a single subcutaneous injection of BUP-XR (300 mg) at different locations, shown for the first 48 h (A) and the entire treatment period (B). Shown for the safety analysis set. BUP-XR, extended-release buprenorphine; SD, standard deviation



4 Discussion

The present study supports BUP-XR administration into the upper arm, buttocks, and thigh in addition to the abdomen. After a single 300-mg BUP-XR injection, buprenorphine plasma exposure ($AUC_{0-28\text{days}}$) was similar across all injection sites, with geometric mean ratios ranging from 95.89 to 114.11%. The mean buprenorphine plasma concentrations in the plateau phase were comparable and sustained at close to or above 2 ng/mL, which is the therapeutic concentration targeted for opioid blockade [12, 13, 15].

A higher peak buprenorphine concentration (C_{max}) was observed after BUP-XR injection into the upper arm and thigh versus the buttocks and abdomen. Together with an earlier T_{max} (median of 4 h and 7 h for the thigh and upper arm, respectively, compared with 10 h and 16 h for the buttock and abdomen), this suggests a more rapid absorption of buprenorphine during the initial depot formation at those sites. This difference could be explained by a lower amount of adipose tissues in the thigh and upper arm compared with the abdomen and buttocks, which could impact the absorption of buprenorphine (lipophilic drug) during the early release process (i.e., at the time of the peak). A previous

Table 2 Pharmacokinetic parameters following a single subcutaneous injection of BUP-XR (300 mg) by injection site group

Pharmacokinetic parameter	Reference	Test location			
	Statistics	Abdomen (<i>N</i> = 22)	Upper arm (<i>N</i> = 20)	Buttocks (<i>N</i> = 22)	Thigh (<i>N</i> = 21)
<i>Buprenorphine</i>					
AUC _{0–28days} (ng·hr/mL)	Geometric LS mean	1640	1720	1570	1870
	GMR (90% CI), %	–	105.04 (81.75–134.97)	95.89 (75.88–121.17)	114.11 (90.05–144.60)
AUC _{last} (ng·hr/mL)	Geometric LS mean	1620	1670	1560	1880
	GMR (90% CI), %	–	102.64 (78.9–133.51)	96.17 (74.40–124.31)	115.47 (89.06–149.71)
C _{max} (ng/mL)	Geometric LS mean	6.61	9.17	7.11	10.1
	GMR (90% CI), %	–	138.71 (103.47–185.94)	107.47 (80.74–143.05)	152.27 (114.01–203.38)
<i>Norbuprenorphine</i>					
AUC _{0–28days} (ng·hr/mL)	Geometric LS mean ^a	619	565	611	584
	GMR (90% CI), %	–	91.19 (61.89–134.36)	98.67 (68.73–141.67)	94.33 (65.70–135.44)
AUC _{last} (ng·hr/mL)	Geometric LS mean	623	558	585	589
	GMR (90% CI), %	89.63 (61.62–130.37)	93.90 (65.14–135.35)	–	94.56 (65.32–136.90)
C _{max} (ng/mL)	Geometric LS mean	2.35	2.34	2.34	2.29
	GMR (90% CI), %	–	99.75 (67.87–146.60)	99.80 (68.54–145.31)	97.73 (66.82–142.95)

The natural log-transformed pharmacokinetic parameters of buprenorphine and norbuprenorphine were compared between each subcutaneous test injection site (buttocks, back of the upper arm, thigh) versus the reference site (abdomen); shown for participants included in the pharmacokinetic evaluable analysis set. Numbers analyzed are noted when different than totals for the pharmacokinetic evaluable test set.

AUC, area under the plasma concentration–time curve; BUP-XR, extended-release buprenorphine; CI, confidence interval; C_{max}, maximum plasma concentration; GMR, geometric mean ratio; LS, least squares

^aFor norbuprenorphine AUC_{0–28days}, the numbers evaluated were as follows: *n* = 19, *n* = 16, *n* = 21, and *n* = 21 for abdomen, upper arm, buttocks, and thigh, respectively

population pharmacokinetic analysis of buprenorphine conducted on BUP-XR data from 570 patients did identify BMI as a statistically significant covariate on the rapid buprenorphine absorption rate constant associated with the early buprenorphine peak [16]. More specifically, patients with a smaller BMI had a more rapid absorption of buprenorphine in the early phase of the kinetics. However, the effects of BMI were not of sufficient magnitude to suggest that dose adjustments would be necessary [12]. It is noteworthy that in the current study, buprenorphine peak concentrations were largely overlapping across injection site groups (Supplementary Fig. 4). Only a few participants had higher C_{max} values (in the range of 20–40 ng/mL) following BUP-XR injection in the thigh and upper arm. In total, six of those participants had a BMI value on the lower end of the BMI range in the study (18.3–23.4 kg/m²), consistent with previous

knowledge [16]. Overall, mean plasma concentration–time profiles show that the difference in peak exposure was limited to the first 6 h of the kinetics, after which buprenorphine plasma concentrations were comparable across all injection site groups. Importantly, buprenorphine levels during the plateau phase were similar and maintained at therapeutic target levels, supporting similar efficacy. This suggests that when the subcutaneous depot is formed, the slow release of buprenorphine from the depot becomes the rate limiting factor and any difference in adipose tissue composition has no or limited impact on buprenorphine absorption.

From a safety perspective, there were no unexpected findings. BUP-XR injection into alternative sites was well tolerated, with safety profiles comparable to that of injection in the abdomen in this study and prior studies [14, 20]. The increased buprenorphine C_{max} after BUP-XR injection into

Table 3 Treatment-emergent adverse events following BUP-XR injection

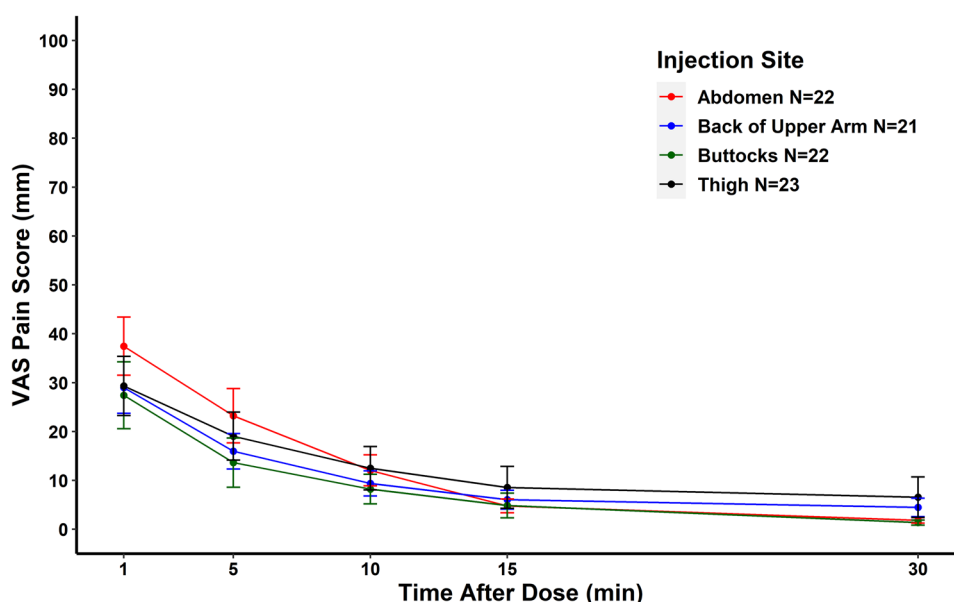
	Reference	Test location			Overall
	Abdomen (N = 22)	Upper arm (N = 21)	Buttocks (N = 22)	Thigh (N = 23)	Total (N = 88)
Any AE	5 (22.7)	3 (14.3)	4 (18.2)	2 (8.7)	14 (15.9)
Risk difference compared with abdomen (95% CI) ^a		– 8.44 (– 31.48, 14.59)	– 4.55 (– 28.34, 19.25)	– 14.03 (– 34.99, 6.93)	
Serious AEs	0	0	0	0	0
Injection site reaction AE	2 (9.1)	1 (4.8)	1 (4.5)	1 (4.3)	5 (5.7)
Risk difference compared with abdomen (95% CI) ^a	–	– 4.33 (– 19.40, 10.75)	– 4.55 (– 19.38, 10.29)	– 4.74 (– 19.36, 9.88)	–
AE related to BUP-XR	2 (9.1)	1 (4.8)	3 (13.6)	1 (4.3)	7 (8.0)
AE reported as opioid withdrawal symptom	0	0	0	0	0

Incidence shown as *n* (%) for the safety analysis set. Risk differences shown as %

AE, adverse event; BUP-XR, extended-release buprenorphine; CI, confidence interval; AE, treatment-emergent adverse event

^aA positive risk difference suggests higher risk in the non-abdomen injection site. Miettinen–Nurminen confidence interval is used

Fig. 3 Mean (+ SEM) visual analog scale pain score (mm) after BUP-XR injection at different locations. SEM, standard error of the mean



the upper arm or thigh was not associated with an increased incidence of AEs or changes in any other measures of injection tolerability. Overall, relatively few participants had injection site AEs (5/88, 5.7%) and no individual injection site reaction was reported for more than 2 participants overall. No one reported injection site pain, tenderness, erythema/redness, or induration/swelling of “severe” intensity, and all measures were graded “mild” or “none” at 2 h post-injection. Mean injection site pain VAS scores and the percentage of positive responses to the burning/stinging question decreased over time, with no differences between injection site groups.

While this study evaluates a single dose of BUP-XR, no major safety concerns would be expected with monthly

dosing. Buprenorphine is considered extremely safe in comparison with full agonist opioids due to its partial agonist activity at mu-opioid receptors and ceiling effect on respiratory parameters [9, 24, 25]. Therefore, while buprenorphine plasma levels will increase with repeated dosing (reaching an average concentration of 6.3 ng/mL and peak concentration of 11.8 ng/mL with 300 mg monthly at steady state [12]), a moderate and transient increase in peak plasma concentration up to 52% (17.9 ng/mL) is not expected to impact patients’ safety. In Australia, SL buprenorphine has been approved for the treatment of moderate-to-severe OUD at doses up to 32 mg daily, which delivers a buprenorphine C_{max} of 15.6 ng/mL [26]. Umbricht et al. [27] showed that buprenorphine was safe even after intravenous

administration. In a study in which experienced opioid users received escalating doses of 2, 4, 8, 12, and 16 mg of buprenorphine intravenously, dose-response curves showed no clear relationship with the dose and suggested a ceiling effect on cardiorespiratory and subjective measures [27]. The mean maximum decrease in respiratory rate was similar across all dose levels and was not associated with clinically significant decreases in oxygen saturation. Buprenorphine $C_{\max}$ values measured in this study ranged from 19.3 ± 1.0 ng/mL (mean $\pm$ standard error) at 2 mg to 124.6 ± 16.6 at 12 mg and 137.7 ± 18.8 ng/mL at 16 mg [28]. Peak concentrations observed after intravenous administration of 12 mg and 16 mg buprenorphine were much higher than the peak concentrations reached with 300 mg BUP-XR at steady-state (11.8 ng/mL), even after accounting for a potential increase up to 52%. The safety of 300 mg BUP-XR monthly is supported by long-term safety data covering up to 18 months of treatment [19, 20].

There were limitations to this study. This study was designed as a parallel group study, because a crossover design was not feasible with this long-acting formulation. In addition, the study was not blinded, which could lead to potential assessment bias. However, the primary endpoints, defined as buprenorphine overall and peak exposures (measured by $C_{\max}$ and $AUC_{0-28 \text{ days}}$), were objective measures that may be less impacted by potential bias compared with other safety or efficacy assessments. Furthermore, safety and tolerability measures included spontaneous adverse event reporting, patient-reported outcome scales and medical staff assessments, and comparative results across these differing modalities remained consistent.

5 Conclusions

The pharmacokinetic and safety findings from this study support the use of the upper arm, thigh, and buttocks in addition to the abdomen for administration of BUP-XR in patients with moderate-to-severe OUD. BUP-XR achieved and sustained therapeutic buprenorphine plasma concentrations over the monthly dosing interval for all injection sites, with comparable buprenorphine plasma exposure and no clinically meaningful differences in safety or tolerability. Offering alternative injection sites is expected to provide additional flexibility to patients, potentially improving adherence and satisfaction, particularly for those who may prefer less visible or more convenient injection locations. This flexibility is crucial given the chronic nature of the disease and the need for long-term sustainable treatment strategies.

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Declarations

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Conflicts of interest C.M.L., D.M., and R.D. are employees of Indivior Inc. O.L. received support from Indivior Inc. to conduct the study. The study was funded by Indivior Inc.

Ethics approval The study was approved by an independent Institutional Review Board (Advarra, Inc., Columbia, MD, USA) on 19 December 2022 (approval no.: Pro00068079).

Consent for publication Not applicable.

Informed consent All participants provided written informed consent before any protocol-related procedures commenced.

Availability of data and material Data are available from the authors upon reasonable request. All requests for raw and analyzed data will be promptly reviewed by the sponsor delegate to verify whether the request is subject to any confidentiality obligations. Patient-related data not included in the paper were generated as part of clinical trials and may be subject to patient confidentiality. Any data that can be shared will be released via a data use agreement.

Code availability Not applicable.

Author contributions C.M.L., D.M., and R.D. contributed to the study design and data analysis. O.L. conducted the study. All authors were involved in the interpretation of the data and critically reviewed the manuscript.

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